

ReadMe for Replicating

A Bayesian Alternative to Synthetic Control for Comparative Case Studies¹

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Computational platforms:

1. We tested our code on an iMac5K 2020 with Intel 6-core processor and 64G RAM. The operation system is MacOS 10.15 (Catalina).
2. The R version is 3.6.1.

Installing packages

1. Please install packages “haven” (>2.2.0), “ggplot2” (>3.3.3), “doParallel” (>1.0.16), “foreach” (>1.5.1), “abind” (>1.4.5), “xtable” (>1.8.4), “coda” (>0.19.4), and “Synth” (>1.1.5) from CRAN.

2. Please install “gsynth” (1.1.7), “panelView” (1.1.3) and “pblasso” (1.0.8) from tar.gz files from the “packages” folder, for example, in Mac/Linux command lines:

```
>R CMD install gsynth_1.1.7.tar.gz
```

or install from files in RStudio.

Please note that “gsynth” has following dependent packages: “GGally” (>2.2.1), “mvtnorm” (>1.0.12), “lfe” (>2.8.6), “Rcpp” (>1.0.6), “RcppArmadillo” (>0.10.2). “panelView” is dependent on “gridExtra” (>2.3). Please install them via CRAN.

3. Please check [this page](#) if you encounter an installation error, such as the “-lgfortran” error. Please also check “Solution_Install_Error.pdf”. At this moment, we do not provide support for MacOS Big Sur.

Folders and their functionalities

code	Stores code files required to simulate and plot data, and summarize results
data	Stores data from ADH (2015)
graphs	Intended for storing graphs from code files
packages	Stores required packages
tempdata	Stores intermediate empirical and simulation results (>800M)

Notes

1. Please make sure to set path to the root replication folder in R.
2. Please execute the code files sequentially based on the name order.
3. All graphics are saved in the “graphs” folder. When you run the code, existing files in these folders will be overwritten, which is normal.
4. Figure 1, Tables 1 and 2 present information on the model setup and comparisons with other methods. Replication is not applicable.

¹ Please email Yiqing Xu (yiqingxu@stanford.edu) if you have any questions related to replication.

5. Certain results (e.g., runtime and credibility intervals) may not be exactly republished due to randomness of the simulation processes.
6. Full replication takes a few days to finish. To speed things up, you can download the intermediate files that store empirical and simulation results [here](#) (821M). Please unzip the file and put it in the root directory of the replication folder. They are not updated to *Dataverse* because of the large file sizes.

Mapping file names to Figures appearing in the paper

In Paper	File Name	In Paper	File Name
Figure 2(a)	sim0_att0.pdf	Figure A1	adh_outcome.pdf
Figure 2(b)	sim0_att.pdf	Figure A2	adh_trace1.pdf
Figure 3(a)	sim0_beta.pdf	Figure A3	adh_xi.pdf
Figure 3(b)	sim0_xi.pdf	Figure A4	adh_wg.pdf
Figure 3(c)	sim0_wg.pdf	Figure A5	adh_fl.pdf
Figure 4(a)	sim_baseline_rmse.pdf	Figure A6	xu_treat.pdf
Figure 4(b)	sim_baseline_cover.pdf	Figure A7	xu_outcome.pdf
Figure 5(a)	adh_ct0.pdf	Figure A8	xu_trace1.pdf
Figure 5(b)	adh_ct.pdf	Figure A9	xu_wg.pdf
Figure 6(a)	adh_att0.pdf	Figure A10	xu_fl.pdf
Figure 6(b)	adh_att.pdf	Figure A11(a)	xu_iattg1.pdf-xu_iattg6.pdf
Figure 6(c)	adh_att_p.pdf	Figure A11(b)	xu_iatt1.pdf-xu_iatt6.pdf
Figure 7(a)	xu_att2.pdf.	Figure A12	sim0_treat.pdf
Figure 7(b)	xu_att1.pdf	Figure A13	sim0_outcome.pdf
		Figure A14	sim0_alpha.pdf
		Figure A15(a)	sim0_trace1.pdf
		Figure A15(b)	sim0_trace1.pdf
		Figure A16(a)	sim_single_rmse.pdf
		Figure A16(b)	sim_single_cover.pdf

Code files

Filename	Description	Uses	Output	Execution Time
0_master.R	Execute all code files.			
1_ex_sim0.R	Performs analysis on a simulated dataset.		- Figures 2, 3, A12-A15, - sim0.RData	Minutes
2_ex_adh2015.R	Performs analysis on the German reunification example from ADH (2015).		- Figures 5, 6, A1-A5 - adh2015.RData	Minutes
3_ex_xu2017.R	Performs analysis on the EDR example from Xu (2017).		- Figures 7, A6-A11 - xu2017.RData	Minutes
4_sim_baseline.R	Conducts baseline simulations to investigate the role of each component of the DM-LFM.		sim_baseline.RData	Hours, possibly days
5_sim_gamma.R	Conducts simulations to investigate whether non-normally distributed error affects model performance		sim_gamma.RData	Hours
6_sim_normal.R	Conducts simulations to investigate whether normally distributed factors affect model performance.		sim_normalF.RData	Hours
7_sim_shortpanel.R	Conduct simulations to investigate whether a small T0 or too few control units affect model performance.		sim_shortpanel.RData	Hours
8_sim_single_r3.R	Conduct simulations to investigate cases with a single treated unit with three strong factors.		sim_single_r3.RData	Hours
9_sim_single_r8.R	Conduct simulations to investigate cases with a single treated unit with eight weak factors.		sim_single_r8.RData	Hours
10_sim_single_X.R	Conduct simulations to investigate cases with a single treated unit with both factors and covariates.		sim_single_X.RData	Hours
11_summ_baseline.R		- sim_baseline.RData - sim_gamma.RData - sim_normalF.RData - sim_shortpanel.RData	- Figure 4 - Tables A1-A4	Minutes
12_summ_single.R		- sim_single_r3.RData - sim_single_r8.RData - sim_single_X.RData	Tables A5-A7	Minutes